Elements of AIML
Miscellaneous – Lecture 01 Notes
PCA, LDA, and ICA in One View

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1 Lecture Overview

This supplementary lecture puts **PCA**, **LDA**, and **ICA** side by side. Students often remember their names but confuse:

- what each method optimizes,
- whether labels are needed,
- what kind of output each method produces,
- when one method should be preferred over another.

The aim here is not to replace the Unit III lecture, but to make the comparison easier to revise later.

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.

2 How to use these notes

Recommended order:

1. First read **Unit 03 Lecture 03** for PCA basics and the leakage-safe preprocessing rule.
2. Read this lecture to compare PCA, LDA, and ICA more systematically.
3. Read **Miscellaneous Lecture 02** if you want the full manual eigenvalue/eigenvector derivation behind PCA.

3 What should already be familiar?

- From **Unit 03 Lecture 03**: covariance matrix, eigenvalues, eigenvectors, and explained variance.
- From **Unit 04**: class labels, classes, and why classification quality depends on separability.

If these ideas are weak, revisit those lectures first; otherwise the comparison here will feel more abstract than useful.

4 A common confusion: same category, different goals

PCA, LDA, and ICA are all discussed under **dimensionality reduction** or **feature transformation**, but they do not solve the same problem.

4.1 A useful first question

Before choosing one of the three methods, ask:

1. Do I have labels?
2. Do I want compression, class separation, or latent-source recovery?
3. Do I care more about variance, discrimination, or independence?

5 PCA

5.1 Core idea

Principal Component Analysis (PCA) finds orthogonal directions that capture the largest variance in the data. It is unsupervised: labels are not used.

5.2 What PCA optimizes

If C is the covariance matrix, PCA chooses a unit vector v that maximizes projected variance:

$$\max_{\|v\|=1} v^\top C v$$

The first solution is the eigenvector corresponding to the largest eigenvalue. The second solution is orthogonal to the first, and so on.

5.3 What PCA gives back

- principal directions (eigenvectors),
- variance captured by each direction (eigenvalues),
- transformed coordinates in the reduced space.

5.4 When PCA is useful

- data compression,
- noise reduction,
- 2D or 3D visualization,
- reducing redundancy in correlated features.

5.5 Where PCA can disappoint

- High variance is not always the same as high usefulness.
- PCA may keep directions that are statistically strong but not class-discriminative.
- Components are often harder to interpret than original features.

6 LDA

6.1 Core idea

Linear Discriminant Analysis (LDA) tries to find projections that separate classes well. Unlike PCA, it is supervised because it uses class labels.

6.2 The two scatter matrices

LDA balances:

- **within-class scatter** S_W: how spread each class is inside itself,
- **between-class scatter** S_B: how far class means are from each other.

The usual criterion is:

$$J(W) = \frac{|W^\top S_B W|}{|W^\top S_W W|}$$

So the method wants:

- large separation between class centers,
- small spread within each class.

6.3 What LDA gives back

LDA returns discriminant directions. These directions are chosen for class separation, not for overall data variance.

6.4 Why LDA has a dimension limit

If there are C classes, LDA can return at most $C - 1$ useful directions. This happens because between-class information depends on class means, and there are only $C - 1$ independent mean-separation directions.

6.5 When LDA is useful

- when class labels are available,
- when the purpose is classification support,
- when you want a lower-dimensional representation that still preserves class discrimination.

6.6 Where LDA can disappoint

- If classes overlap strongly, LDA cannot force clean separation.
- If labels are noisy, the projection becomes less meaningful.
- Without labels, LDA cannot be applied at all.

7 ICA

7.1 Core idea

Independent Component Analysis (ICA) assumes observed features are mixtures of hidden latent sources. The aim is to recover components that are as statistically independent as possible.

7.2 Mixing model

ICA is usually explained by:

$$X = AS$$

where:

- S contains hidden source signals,
- A is an unknown mixing matrix,
- X is what we observe.

The job of ICA is to estimate an unmixing matrix so that the recovered components behave independently.

7.3 Why ICA is different from PCA

PCA only removes correlation and ranks directions by variance. ICA goes further: it tries to recover statistically independent components, which is a stronger condition than uncorrelatedness.

7.4 Typical ICA pipeline

1. center the data,
2. often whiten the data,
3. optimize an independence criterion, often related to non-Gaussianity.

7.5 When ICA is useful

- signal separation,
- EEG or audio artifact removal,
- recovering hidden source-like factors from mixed observations.

7.6 Where ICA can disappoint

- If the latent sources are not independent, the model assumption breaks.
- If the data is close to Gaussian, ICA becomes much less informative.
- ICA components are often sensitive to scaling and preprocessing choices.

8 Detailed comparison

8.1 Comparison by objective

Method	Main objective	Interpretation
PCA	maximize variance	keep the strongest spread directions
LDA	maximize between-class separation relative to within-class spread	help classes separate in a lower-dimensional space
ICA	maximize statistical independence	recover hidden source-like signals

8.2 Comparison by supervision

Method	Uses labels?	Why this matters
PCA	No	cannot use label information, so class separation may be accidental rather than intentional
LDA	Yes	uses labels directly, so it can align the representation with classification needs
ICA	No	focuses on independence structure, not on labels or class boundaries

8.3 Comparison by assumptions

- **PCA**: important information is reflected in large-variance directions.
- **LDA**: classes are meaningful and separability matters.
- **ICA**: latent sources are independent and often non-Gaussian.

8.4 Comparison by geometry/statistics

- **PCA** produces orthogonal components.
- **LDA** produces class-discriminative directions, not necessarily variance-dominant ones.
- **ICA** seeks independence, which is stronger than orthogonality and stronger than simple uncorrelatedness.

8.5 Comparison by output dimension

- **PCA**: any $k \leq d$ may be chosen.
- **LDA**: at most $C - 1$ directions for C classes.
- **ICA**: usually chosen according to the assumed number of sources or observed variables.

8.6 Comparison by classroom use-case

- If you want to compress features before plotting or faster modeling, start with **PCA**.
- If your real goal is classification and labels are available, **LDA** may be more aligned with the task.

- If you are separating mixed signals or hidden factors, **ICA** is the more natural tool.

9 One-sentence memory anchors

- **PCA:** "keep the directions with the most variance."
- **LDA:** "keep the directions that separate classes."
- **ICA:** "recover independent hidden sources from mixtures."

10 What students often say incorrectly

Incorrect statement 1: "LDA is supervised PCA."

Correction: the two methods are related only at a high level as linear transformations. Their objectives are different: PCA maximizes variance, while LDA maximizes discriminative separation.

Incorrect statement 2: "ICA gives more important principal components."

Correction: ICA does not rank components by variance at all. It looks for independence, not for variance dominance.

Incorrect statement 3: "Orthogonal means independent."

Correction: orthogonal vectors are geometrically perpendicular; independence is a stronger probabilistic statement. Two components can be uncorrelated but still not independent.

11 Decision checklist

Use this quick decision process:

1. If labels are absent, remove LDA from the candidate list.
2. If the goal is compression or visualization, prefer PCA.
3. If the goal is better class separation, investigate LDA.
4. If the goal is source separation or mixed-signal recovery, investigate ICA.
5. Check whether the method assumptions actually make sense for the data.

12 Practice Questions (with answers)

Q1. Why can PCA fail to help classification even when it keeps most variance?

Answer: because the directions that contain the most variance may not be the directions that best separate classes. Variance and discriminative power are different objectives.

Q2. Why can LDA not be used without labels?

Answer: because LDA needs class means and class-wise scatter. Without labels, those class-based quantities are undefined.

Q3. Why is ICA more demanding than PCA?

Answer: PCA only uses second-order structure such as covariance, while ICA tries to recover statistically independent latent sources, which requires stronger assumptions and harder optimization.

Q4. Which method would you try first for a 2D visualization of high-dimensional embeddings?

Answer: PCA, because its goal is compression/variance preservation and it is easy to apply without labels.

13 Summary

PCA, LDA, and ICA belong to the same broad family of linear transformations, but they answer different questions:

- **PCA:** what directions preserve the most variance?
- **LDA:** what directions separate known classes best?
- **ICA:** what hidden components are most independent?

The method should therefore be chosen by **goal and assumptions**, not by familiarity of name.

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.